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Multimodal Self- and Other-Initiated Repairs in L2 Peer Interactions

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Abstract

This study examines the multimodal organization of self- and other-initiated repairs in L2 peer interactions between French secondary pupils. While self-repairs typically reflect errors and language difficulties at different levels (syntactic, lexical, phonological, etc.) and are treated as (dis)fluency features, repair mechanisms are not solely responses to linguistic errors but may be used as tools to negotiate meaning and manage misunderstandings, especially through other-repairs. Adopting a multimodal approach, this study focuses on the visual-gestural properties of repairs, based on different articulators (face, head, and hands). Results show a preference for self-repairs over other-repairs, especially those associated with lexical and syntactic difficulties. Other-repairs mainly consist of restricted offers in contexts of misunderstanding and acceptability problems. Findings also reveal distinct multimodal patterns depending on the type of the repair. A more detailed qualitative analysis of the data highlights the sequential relationship between self- and other- repairs.

Index Terms: repair, fluency, multimodality, second language acquisition, interactional competence

1. Introduction

Repairs have been treated and analyzed quite differently in the literature depending on the theoretical framework adopted. In (dis)fluency research for instance, self-repairs are considered a type of (dis)fluency feature which result from a detection of trouble in speech, and aim at repairing a previous 'unwanted' element in the original delivery [1]. Repairs can operate at different levels, namely lexical, syntactic, phonological, and others, depending on the trouble source. Utterances containing repairs typically follow a similar surface structure: the *reparandum*, the *interregnum*, and the *reparans* [2]. In second language research more specifically, repairs point to a distinct aspect of fluency, known as 'repair fluency' [3] which relates to speech repetition, reformulation and correction. In sum, repairs are intricately related to stages of speech production, hence focusing exclusively on the production of an individual speaker, without much taking into account the interlocutor's contribution or interactional dynamics.

Besides the field of (dis)fluency, repair is a key area of research in Conversation Analysis (CA). It refers to an organized system which is addressed to deal with recurrent problems in speaking, hearing, and understanding. Repairs can be distinguished between self- and other-initiated repairs, and while Levelt [1] also initially distinguished between self- and other- types, the latter are usually excluded from (dis)fluency analysis. Unlike what Levelt and other researchers addressed in their models, in CA, repair is "neither contingent upon error, nor limited to replacement" [4]. Repair mechanisms have a specific organization within the course of interaction and can occur within the same turn, but also in a turn's transition space and in a subsequent turn. Other-initiated repairs can be categorized into different subtypes, such as open repairs, restricted request or restricted offers [5, 6] and deal with several interactional problems such as problems of hearing, reference, understanding, and acceptability [7].

Besides speech and talk, the *multimodal* nature of repairs is also of great value [8]. Repair sequences can be accompanied by a variety of kinetic and bodily resources, such as hand gestures [9, 10], gaze practices [11], head movements [12] as well as body movements [13]. Therefore, the aim of the present study is to analyze self- and other-initiated repairs in L2 peer interactions with a multimodal approach, combining conversation-analytic methods with corpus-based annotation models implemented in (dis)fluency research. While CA-oriented studies offer richly detailed and careful examinations of repairs in their sequential and interactive environment, annotation models provide a broader picture of the phenomena under study through systematic coding. In line with mixed-methods [14] these approaches should thus be considered complementary rather than contradictory. In particular, this methodology is applied to L2 data, where repairs may relate to the notion of interactional competence, as described in the following section.

2. Previous work on repairs in L2

In the context of Second Language Acquisition (SLA), and more specifically L2 peer interactions, repair sequences play a crucial role not only in addressing linguistic inaccuracies but ensuring mutual understanding as well as maintaining and organizing the flow of interaction. This also relates to the concept of *interactional competence* [15,16] – the ability to manage and participate effectively in conversational exchanges. Recent research in SLA has shown how peer learners often use both self- and other-initiated repairs to address misunderstandings or knowledge gaps to negotiate meaning, monitor understanding, and build knowledge [17,18,29]. While self-repairs are typically linked to *individual fluency* management, including self-monitoring practices such as lexical retrieval, reformulation and syntactic reorganization [19, 20], other-repairs deal with *interactional fluency*, in contexts of joint, collaborative word searches [21, 22]. In any case, self- and other-repair are intricately related, as other-repairs can be prompted by self-repairs in contexts of completion [4] or misunderstanding. As this paper will show, self-initiated repairs caused by language difficulties in the L2 may give rise to other-repairs due to a lack of understanding, where the trouble source is resolved collaboratively.

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Self-repairs in particular also have a number of temporal properties: while they typically take the form of repetitions, cut-offs, replacements, and reformulations [23], they also include a time interval, also called an *editing phase* [1], during which speakers insert pauses (silent or filled) or editing terms, such as 'I mean, I think' and the like. As shown by [24], the duration of the editing phase may be influenced by the type of self-repair, depending on the trouble source (appropriateness repair versus error repair). In addition, a growing body of research in SLA has been paying attention to the multimodal organization of repairs, where visual-gestural resources contribute to the initiation and resolution of trouble. One study conducted on language instructors during L2 Chinese classroom interactions has shown how eyebrow raises and head movements function as visual-only repair initiators, subtly prompting learners to self-monitoring without explicit verbal cues [25]. Similarly, another study conducted on classroom interaction highlighted the different semiotic resources mobilized by instructors to initiate or enact a repair, such as cut-off body movements, or "motionless gaze" [26]. The "body torque" configuration, consisting of a divergent orientation of the body, also typically arises during other-repairs, displaying increased interactional involvement and thus a preference for intersubjectivity [27].

In sum, repairs have sequential, temporal, as well as visual-gestural properties, and arise in speech and interaction for a number of reasons, both cognitive and interactional, depending on the trouble source. As reported by CA studies [4], self-repairs are more frequently initiated than other-repairs, showing a preference for self-correction over potential social friction through other-repair. This was also found to be the case in the data under study which includes L2 interactions between secondary school pupils [28]. In a previous study conducted on the same data, self-repairs were found to be very frequent, especially lexical and morphosyntactic ones [29], coupled with a recurrent use of pauses. The present research expands on this work by including more detailed visual-gestural analyses. In particular, the research questions (RQs) are the following: (1) Is there a preference for self- or other-repairs in L2 peer interactions? (2) What are the most recurrent types of self-repairs? (3) What are the temporal properties of self-repairs and are they associated to a certain subtype or trouble source? (4) What are the visual-gestural properties of repairs, based on the type (self- versus other) and does the type yield different patterns?

3. Method

The study is based on a selection of video-recorded conversations between lower intermediate secondary school pupils (aged 13-16 years old) engaged in an English-speaking task involving everyday topics (family, hobbies, future career plans, etc.). The corpus was collected as part of a broader longitudinal study [28] involving 48 pupils from 4 different classrooms across 3 secondary schools in France recorded at the beginning and end of the school year. The present sample includes 12 pupils (6 pairs) from one class during the first recording session (see Table 1).

This research is also a follow-up study on the work conducted by [29] on repairs on a larger sample of the data, which was done independently, with differences in methodology. A placement test was administered at the beginning of the school year by the instructors to assess students' proficiency levels. The test included multiple-choice questions covering syntax, vocabulary, and phonology, along with reading comprehension and two short writing tasks. Scores out of 20 were obtained and divided into four proficiency levels: A (16-20) B (12-15) C (8-11) and D (⩽7). The participants' labels correspond to these proficiency levels.

Table 1: *Sample under study*

Pair number	Participants' codes	Speaking time
13	A09 - A10	3.6 - 3.4 mn
14	B09 - B10	3.3 - 3.2 mn
15	C11 - C12	2.7 - 2.8 mn
16	A12 - B12	2.6 - 2.53 mn
17	A11 - C10	1.4 - 2.8 mn
18	B11 - C09	3.03 - 1.9 mn

Repairs were categorized into two main types: self- and other-repairs. Self-repairs were further categorized in different subtypes, following [23]: (1) false start (cut-off word or sentence), (2) repetition (of the same word or sentence), (3) replacement (of a single word or group of words) and (4) reformulation (of a multiple-word unit). The different trouble sources were also identified, following [2,24] including: lexical, (morpho-)syntactic, phonological, and appropriateness. Appropriateness repairs are usually distinguished from error repairs as they are felt to be "undesirable" by the speaker, but not linked to grammatical difficulties. Lastly, the presence of an editing phase was also coded, distinguishing between silent pauses (SP) and filler particles (FP) [30]. For the latter, the duration (in ms) and form (euh, eum, or mm) was also annotated. When both were present, it was coded as "FP+SP".

For other-repairs, they were coded as being either (1) open – requesting clarification or signaling misunderstanding without specifying the source of the problem (e.g. "uh?" "what?") (2) restricted request - seeking specification towards a specific element, and (3) restricted offer – seeking confirmation and offering a candidate understanding [5, 6]. The trouble sources were also identified, distinguishing between problems of hearing (not hearing correctly), reference (not recognizing the reference), understanding (not understanding the meaning), and acceptability/expectation (clashes in knowledge or expectations) [31]. Another category was added, "word search assistance", when the repair did not strictly deal with a trouble on the recipient's part, but was rather used as a completion of the previous speaker's utterance in a word-searching context.

Visual-gestural features of both repair types were also analyzed, based on different body articulators: (1) gaze direction, averted, towards interlocutor, eyes closed, towards paper; (2) facial expression: neutral, frown, smile, raised eyebrows, pout; (3) gesture: adaptor, referential gesture, pragmatic gesture, no gesture; (4) head/body: tilt, shake, nod, move sideways body, move forward or backwards. For the self-repairs, these features were annotated during the *reparans* more specifically, i.e. the repaired item.

4. Results

The data yielded a total number of 167 repairs, including 119 self-oriented (SR) and 48 other-oriented (OR) repairs, as shown in the boxplots in Figure 1. Despite a strong preference for SR overall, some individual differences can be noted, as in Figure 2, which shows the number of repairs per participants. A chi-square test of independence showed a statistically significant difference, $\chi^2(11, N = 167) = 26.11$, $p = .006$, indicating that the distribution of repair types varied significantly across individuals.

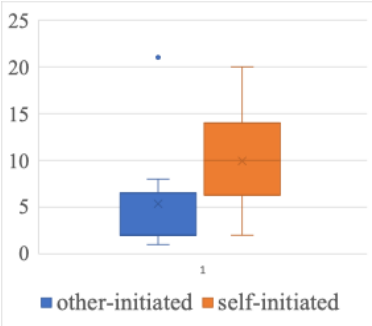

Figure 1: *Self- and other repairs (raw numbers)*

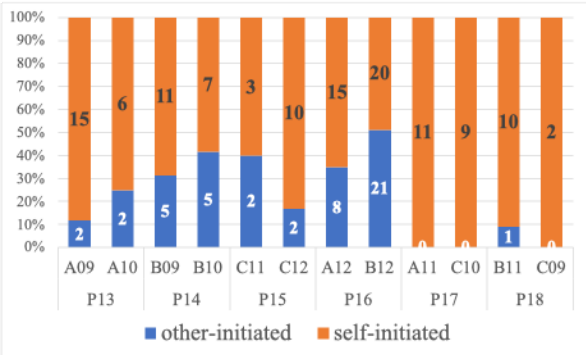

Figure 2: *Repair types per participant (raw numbers)*

For SR types, repetitions and replacements were the most frequent markers (46 and 29 respectively), followed by false starts and reformulations (26 and 18 respectively). A Chi-square Goodness-of-Fit test ($\chi2(3)=14.96$, $p <.05$) showed a significant difference, with repetitions occurring more frequently than expected (critical value = 7.815). Another Chi-square test revealed a significant difference in the distribution of self-repairs across the four types of trouble source (appropriateness, lexical, phonological, and syntactic, $\chi^2(3)=23.15$, $p < .05$) with lexical and syntactic repairs (46 and 38 respectively) occurring more frequently than the other types. If we further cross self-repair type with source (Table 2), a significant association can be found between the two, ($\chi^2(9) = 41.01$, $p < .001$, Cramer's V = .34, indicating a moderate-to-strong relationship. Lexical issues were frequently resolved through repetitions, phonological issues through false starts, and syntactic issues through reformulations.

Table 2: *Self-repairs and trouble source* (numbers)

	App.	Lexical	Syntactic	Phon.
FS	2	4	8	12
REF	4	5	9	0
REP	5	26	14	1
RPC	6	11	8	3

98 out of the 119 self-repairs contained an editing phase, either a filler particle (N= 19), a silent pause (N= 46) or a combination of the two (N= 33). Silent pauses (N= 73 in total, whether isolated or combined) lasted on average 844 ms (SD = 573) and filler particles (N= 37, 24 *euh*, 12 *eum*, and 1 *mm*) had an average duration of 522 ms (SD = 571). The analysis of the association between the self-repair type and the presence of an editing phase (Table 3) reveals that repetitions tend to be more frequently associated with FP+SP combinations ($p = .03741$, $\chi^2 = 41.0139$). However, the duration of pauses does not differ across repair types, as evidenced by the non-significant results from the Kruskal-Wallis tests for both filler particles ($p = .08$) and silent pauses ($p = .19$). This could suggest that pause length is more speaker-dependent than repair-type dependent in the data under study.

Table 3: *Self-repairs and temporal features (numbers)*

	FP	SP	FP+SP
FS	3	11	4
REF	3	10	3
REP	7	12	**20**
RPC	6	13	6

For OR (N= 49), the Chi-square test revealed a significant difference between the different repair types ($\chi^2 = 6.58$, $p = 0.0135$), with restricted offers occurring more frequently (N=26) than both open repairs (N=12) and restricted requests (N= 11) (critical value = 5.599). Findings also showed that a majority of OR dealt with issues in understanding (N=20/48) but also acceptability (N=20/48). In addition, as shown on Table 4, OR were found to be more frequently associated with issues in understanding, while restricted offers and restricted requests were more closely linked to acceptability problems ($\chi^2(8) = 22.83$, $p = .0036$, Cramer's V = .49). This very much aligns with previous findings in the literature [4,5,6,10]

Table 4: *Other-repairs and trouble source (numbers)*

	Open	R. offer	R. request
Acceptability	0	13	6
Hearing	0	4	0
Reference	0	3	0
Understanding	11	4	5
Assistance	0	0	1

Lastly, Table 5 reports all the visual-gestural features for the two repair types, focusing on each articulator (face, eye, hand gestures, and head).

Table 5: *Visual-gestural features* (numbers and %)

Feature	OR	%	SR	%	Total
Facial Expression					
Frown	5	10.4%	3	2.5%	8
Neutral	16	33.3%	69	58.0%	85
Pout	0	0.0%	11	9.2%	11
Raised Eyebrows	10	20.8%	18	15.1%	28
Smile	17	35.4%	18	15.1%	35
Gaze Direction					
Averted	1	2.1%	35	29.4%	36
Eyes Closed	0	0.0%	11	9.2%	11
Paper	3	6.3%	26	21.8%	29
Interlocutor	44	91.7%	47	39.5%	91
Gesture					
Adaptor	8	16.7%	46	38.7%	54
Pragmatic Gesture	4	8.3%	20	16.8%	24
Referential Gesture	0	0.0%	1	0.8%	1
No Gesture	36	75.0%	52	43.7%	88

Feature	OR	%	SR	%	Total
Head Movement					
Head Beat	6	12.5%	21	17.6%	27
Headshake	5	10.4%	6	5.0%	11
Move Backward	2	4.2%	6	5.0%	8
Move Forward	27	56.3%	16	13.4%	43
Move Sideways	3	6.3%	21	17.6%	24
None	5	10.4%	49	41.2%	54

A significant association was found between facial expression and repair type, ($\chi^2(4) = 20.35$, $p = .0004$, Cramer's $V = 0.35$). Smiles were more frequent during OR (35.4%), whereas neutral expressions predominated in SR (58.5%). For gaze direction and repair type, the association was also significant, ($\chi^2(3) = 38.16$, $p < .001$, Cramer's $V = 0.48$). In OR, participants predominantly looked toward the interlocutor (91.7%), while SR were more often accompanied by gaze aversion (29.4%) or looking at paper (21.8%). Eyes closed and averted gaze almost only occurred during SR. A significant association was further found for gesture type, ($\chi^2(3) = 13.59$, $p = .0035$, Cramer's $V = 0.29$), suggesting that certain gestures are more likely to co-occur with either SR or OR (e.g. adaptors during SR). Lastly, there is a significant association between head movement and repair type ($\chi^2 = 39.55$, $p < .001$, Cramer's $V = 0.49$). Notably, moving forward is much more common in OR (56.3%), while SR are most often accompanied by no head movement at all (41.2%).

5. Illustrative case

The following excerpt, taken from one pair of the corpus (P16) further illustrates the sequential relationship between SR and OR. In this extract, B12, who is reading from a list of questions, is having trouble understanding A12's utterances. The transcription conventions are adapted from [32].

```
(1) Best Friend, Pair 16 (01:02)

1 B12:   a:and what's the name of your
+breath- best friend?+
+eyes closed, raised eyebrows+
+head beat-----------+
2 A12:  his name is =
        = °[!]° I have a lot of *best friends*
        °frown°  *repetitive taps with index*
3 A12: +you don't have a best friend?+
        +headshake-----------------+
4 B12   *°I ha-° I have lots*.
        °eyes closed°
        * both open palms facing A12*
        * head beat--------*
5 A12: +you don't have friends?+
        + raised eyebrows------+
6 B12: I have *↑MAny* best friends.
                *head beat*
        *open palm produced in a wave*
7 A12: you °don't° have *friends*?
            °frown°     *moves forward*
8 B12 [!] I have *↑MAny* a [*°lot° of friends*
    *flat hand going down* *pointing gesture*
                           °raised eyebrows°
9 A12                       [*AH↑ many.*
                             *head beat*
```

B12 opens the sequence by reading from the list of topics provided to them, asking A12 if he has a best friend. B12 first produces a phonological repair on the adjective "best", accompanied by raised eyebrows and a head beat. A12's immediate answer is to provide the name of his best friend, but then changes his initial thought with a tongue click accompanied by a frown, and replaces his initial trajectory ("'I have a lot of best friends", l.2). The quantifier 'a lot of', becomes the main trouble source in the subsequent turns, leading to B12's first restricted offer, used for seeking confirmation, followed by two others: "you don't have friends?" (l. 2-7). This prompts A12 to initiate a number of self-repairs, first a false start (l. 4) then a replacement (from " a lot of" to "many", produced twice with prosodic stress). Each time, the self-repairs are co-produced with a series of hand gestures (including beats, as shown by [9]) serving pragmatic functions. The misunderstanding is finally resolved after a second repetition of the quantifier 'many', during which B12 displays his understanding with the interjection 'ah' and a repetition of the word. What this example illustrates is how sequentially coordinated self- and other-repairs are, and thus do not solely relate to individual trouble management. Indeed, the series of SR produced by A12 were prompted by B12's initial OR.

6. Concluding discussion

This study examined the multimodal organization of self-initiated and other-initiated repairs in L2 peer interactions, with a focus on repair type, trouble source, temporal features, and embodied resources. The findings confirm the well-established preference for self-repair over other-repair [4], with learners primarily addressing lexical and syntactic difficulties (RQ1). Among SR types, repetitions and replacements were the most frequent, with repetitions often associated with lexical issues and co-produced with filler particles and silent pauses (RQ2 & 3). This aligns with previous research linking self-repair to fluency regulation and temporal patterns [2, 24]. Findings also revealed that OR occurred mostly as open repairs and restricted offers in contexts of misunderstanding and acceptability trouble. These results echo previous work showing how other-repairs play a role in managing intersubjectivity and alignment [22, 33]. In addition, the qualitative analysis of a data fragment further demonstrated the sequential relationship between SR and OR, with SR triggering OR in the subsequent turns when the latter were not fully resolved, hence highlighting the dynamic and co-constructed nature of repair mechanisms.

Multimodal analyses point to clear differences between repair types (RQ4), with different visual-gestural patterns of co-occurrence. While SR were most often accompanied by gaze aversion and adaptors, OR involved mutual gaze, forward head movement, and smiles. This supports the multimodal nature of repair, as reported in previous studies [8, 9, 10, 34]. Hand gestures, together with gaze practices, facial expressions and head movements are integral to the trajectory of the repair.

Despite its contributions, the present study is limited by the relatively small sample (only 12 participants, yielding 167 tokens), which constrain generalizability. Future work could expand the dataset (in line with [29]) and compare the first two sessions to see if repair practices are likely to change over the span of a school year. More qualitative analyses of data fragments could also be compiled in a case study, to better examine the sequential and multimodal environment of SR and OR. Nevertheless, these preliminary findings offer valuable insights, and highlight the importance of integrating multimodal and interactive approaches to uncover how learners collaboratively manage and build fluency in conversation.

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8. References

[1] W.J. Levelt, "Monitoring and self-repair in speech."*Cognition,* vol. 14, no. 1, pp. 41-104, 1983

[2] B. Pallaud, , R. Bertrand, L. Prevot, P. Blache, and S. Rauzy. "Suspensive and disfluent self interruptions in French language interactions." *Fluency and Disfluency across Languages and Language Varieties. Corpora and Language in Use–* vol. 4, pp. 109-138, 2019

[3] P. Tavakoli, "Assessment of second language fluency." *Language Teaching*, pp. 1-17, 2025.

[4] E.A. Schegloff, G. Jefferson, and H. Sacks. "The preference for self-correction in the organization of repair in conversation." *Language,* vol. 53, no. 2, pp. 361-382, 1977

[5] J. Baranova "Other-initiated repair in Russian." Open Linguistics vol. 1, no. 1, pp. 555-577, 2015

[6] M. Dingemanse, K. H. Kendrick, and N. J. Enfield. "A coding scheme for other-initiated repair across languages." *Open Linguistics*, vol. 2, no. 1, pp. 35-46, 2016.

[7] I. Mostovaia and M. Pfeiffer, Other-initiated self-repair. In A. Gubina, E. M. Hoey and C. W. Raymond (Eds.), *Encyclopedia of Terminology for Conversation Analysis and Interactional Linguistics. International Society for Conversation Analysis (ISCA)* 2023

[8] M. Rasenberg, W. Pouw, A. Özyürek, and M. Dingemanse. "The multimodal nature of communicative efficiency in social interaction." *Scientific Reports,* vol. 12, no. 1. 2022

[9] C. Bellifemine, "Non-referential beat gestures and their function of self-repair in children's narratives." *Journal of Child Language Acquisition and Development-JCLAD, vol.* 12, no. 1 pp. 963-987, 2024

[10] F. Oloff, "“Sorry?”/“Como?”/“Was?”–Open class and embodied repair initiators in international workplace interactions." *Journal of Pragmatics,* vol. 126, pp. 29-51, 2018.

[11] M.M Egbert, "Context-sensitivity in conversation: Eye gaze and the German repair initiator bitte?." *Language in Society,* vol. 25, no. 4, pp. 587-612, 1996

[12] M.S. Seo, and I. Koshik. "A conversation analytic study of gestures that engender repair in ESL conversational tutoring." *Journal of Pragmatics* vol. 42, no. 8 pp. 2219-2239, 2010

[13] G Rasmussen, "Inclined to better understanding—The coordination of talk and ‘leaning forward ’in doing repair." *Journal of Pragmatics, vol.* 65 pp. 30-45. 2014

[14] M.R. Hashemi and E. Babaii. "Mixed methods research: Toward new research designs in applied linguistics." *The Modern Language Journal, vol.* 97, no. 4 pp. 828-852. 2013

[15] E. Galaczi, and L. Taylor. "Interactional competence: Conceptualisations, operationalisations, and outstanding questions." *Language Assessment Quarterly, vol.* 15, no. 3 pp. 219-236. 2018

[16] S. Pekarek Doehler, "Elaborations on L2 interactional competence: The development of L2 grammar-for-interaction." *Classroom Discourse* vol. 9, no. 1 pp. 3-24. 2018

[17] M.H Chen and S. Xueting Ye. "Extending repair in peer interaction: A conversation analytic study." *Frontiers in Psychology,* vol. 13, 2022.

[18] O. Sert, *Social interaction and L2 classroom discourse*. Edinburgh University Press, 2015.

[19] J. Kormos, "Monitoring and self‐repair in L2." *Language learning,* vol. 49, no. 2 pp. 303-342. 1999

[20] P. Tavakoli, and P. Skehan. "Strategic planning, task structure and performance testing." In *Planning and task performance in a second language*, pp. 239-273. John Benjamins Publishing Company, 2008.

[21] M.H. Chen, and S. Xueting Ye. "Extending repair in peer interaction: A conversation analytic study." *Frontiers in Psychology*, vol. 13, 2022

[22] C.E.Brouwer, "Word searches in NNS–NS interaction: Opportunities for language learning?." *The modern language journal,* vol. 87, no. 4 pp. 534-545. 2003

[23] P. Peltonen, S. Olkkonen, M. Szyszka, and P. Lintunen. "L2 repair fluency through the lenses of L1 repair fluency, cognitive fluency, and language anxiety." *Applied Linguistics Review* vol. 16, no. 2 pp. 877-899. 2025

[24] M. Gósy, I. Vogel, and V. Váradi. "Temporal patterns of error repairs and appropriateness repairs in Hungarian." *The Phonetician,* vol. 113. 61–79. 2013

[25] X. Wang and X. Li. "Teachers’ Eyebrow and Head Movements and Repeats as Other-Initiations of Repair in Second-Language Classrooms." *Social Interaction. Video-Based Studies of Human Sociality,* vol. 6, no. 3, 2024.

[26] L. Kääntä, Teacher Turn-Allocation and Repair Practices in Classroom Interaction. University of Jyväskylä. 2010

[27] A. Kamunen, "How to disengage: suspension, body torque, and repair." *Research on Language and Social Interaction* vol. 52, no. 4 pp. 406-426. 2019

[28] P. Manoïlov, Pascale. L’interaction orale entre pairs en classe d’anglais LV2: analyse didactique et linguistique de la construction et du développement des compétences des apprenants. Université Sorbonne Nouvelle, 2017.

[29] P. Manoïlov "La grammaire du discours-en-interaction chez les apprenants de l’anglais L2: le cas des ajustements." *Linx. Revue des linguistes de l’université Paris X Nanterre, vol.* 81, 2020.

[30] M. Belz "Defining filler particles: A phonetic account of the terminology, form, and grammatical classification of “filled pauses”." *Languages*, vol. 8, no. 1, 2023

[31] T. Benjamin and T. Walker. "Managing problems of acceptability through high rise-fall repetitions." *Discourse processes*, vol. 50, no. 2, pp. 107-138. 2013

[32] L. Mondada "Multiple temporalities of language and body in interaction: challenges for transcribing multimodality." *Research on language and social interaction*, vol. 51, no. 1, pp. 85-106. 2018

[33] P. Drew, "‘Open’class repair initiators in response to sequential sources of troubles in conversation." *Journal of pragmatics*, vol. 28, no. 1 pp. 69-101. 1997

[34] J. Holler "Visual bodily signals as core devices for coordinating minds in interaction*." Philosophical Transactions of the Royal Society* vol. 377, no. 1859. 2022